Multimodal Cross-City Universal OD Flow Prediction

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Abstract

This study aims to construct a multimodal cross-city OD flow prediction framework based on GPT-4.1 Standard Edition. It integrates urban planning texts, street view images, and structured data such as POIs and road networks. Through multimodal feature fusion, cross-city knowledge transfer, and dynamic adaptation mechanisms, it generates accurate OD flow matrices for new cities lacking historical data. This framework breaks through the limitations of traditional single-modal approaches and leverages the model's powerful long-context understanding and multimodal reasoning capabilities to capture implicit associations between urban environments and travel behaviors. It realizes cross-city universal prediction, providing an efficient quantitative tool for transportation planning in emerging cities.

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1 Problem Background

Origin-Destination (OD) flow prediction is a crucial component of urban transportation planning; however, traditional methods have some limitations [1, 2, 3]. These methods not only over-rely on single-modal data and fail to capture the multi-dimensional characteristics of the urban built environment fully, but also mostly adopt traditional prediction algorithms such as statistical models and machine learning models, without leveraging the cross-modal understanding and knowledge transfer capabilities of Large Language Models (LLMs). Additionally, models perform poorly in cross-city transfer—models trained on data from one city often fail when applied to new cities lacking historical travel data [1], due to differences in urban form, cultural norms, and infrastructure.

2 Research Objectives

- Construct an OD flow prediction framework based on **GPT-4.1**, integrating textual, image, and structured data of cities.
- Achieve cross-city universality to provide accurate OD flow matrix predictions for new cities lacking historical travel data.
- Verify the advantage of multimodal feature fusion over single-modal methods in capturing the relationship between urban environment and OD flows (optional, depending on the need for comparison).

3 Research Methods

3.1 Data Preparation

Inputs: 1. **Textual modal data**: Urban planning documents.2. **Image modal data**: Satellite images (if available), street-view images, to capture visual features. 3. **Structured data**: POI spatial coordinates, road network topology (including road grades and connectivity, etc.), cross-city calibration indicators (such as road network density, public transportation coverage).

Output: The final expected output of the model is a **cross-city universal OD flow matrix**, i.e., the travel flow between any two regions in the new city. The matrix should reflect the travel intensity between different regions and be adaptable to the built environment characteristics of the new city.

Three ideas for regional division: 1.Division by urban administrative districts.2.Division by Traffic Analysis Zones (TAZ).3.Division by POI clustering areas.

3.2 Model Architecture

GPT-4.1 is selected as the basic multimodal large language model, with a three-stage framework:

1. **Multimodal feature fusion**

The three types of data are encoded separately and then fused. Text data is converted into semantic vectors through the text encoder of **GPT-4.1** (for example, descriptions such as "technology industrial park" are converted into concepts like "high daytime population density" and "commuting origin"); image data is processed to extract features through the model's image encoder; structured data is transformed into spatial relationships. These features are fused through a cross-attention mechanism.

2. **Cross-city knowledge transfer**

Training cities are clustered into three types—"high-density metropolis", "medium-sized city", and "satellite city"—based on built environment characteristics (such as population density, road network density). Universal features are strengthened, enabling the model to focus on mobility laws common across cities.

3. **OD flow generation**

Multimodal data of the new city is input into the pre-trained fusion model to generate the OD matrix.

4 Main Required Resources

Resource Type	Details
Datasets	Text: Urban planning documents from official portals; Images: Satellite images (if available), street-view images; Structured data: Points of interest, road, and census data from open repositories (possibly from OpenStreetMap)
Computational	Graphics cards for model training/inference (lower requirements if using API calls)

5 Timeline

- **Week 1**: Identify data, complete data collection and cleaning, and divide regions.
- **Week 2**: Extract features, quantify indicators, and split training, validation, and test sets.
- **Week 3**: Build the multimodal feature fusion module, and write the introduction and literature review sections.
- **Week 4**: Implement the city clustering algorithm, complete the initial draft of the model framework and training scripts, and write the "Research Methods" section of the paper.
- **Week 5**: Start training on a single city, and fine-tune the cross-city transfer module using the validation set.

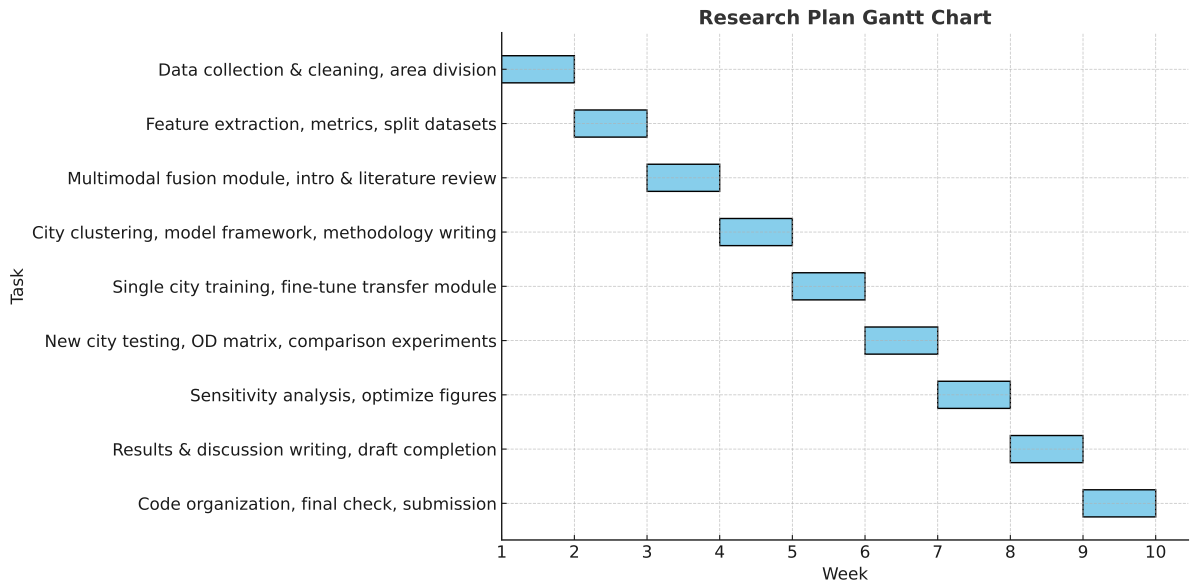

Figure 1: Timeline.

- **Week 6**: Test the model in new cities and generate OD matrices; conduct comparative experiments between single-modal and multimodal models (if applicable).
- **Week 7**: Supplement sensitivity analysis experiments and optimize figures and tables.
- **Week 8**: Write the "Experimental Results and Analysis" and "Discussion" sections of the paper, complete the first draft of the full text, and proofread it.
- **Week 9**: Organize the code repository, conduct repeated checks, and submit the paper.

References

[1] T. He, J. Bao, R. Li, S. Ruan, Y. Li, L. Song, H. He, and Y. Zheng, "What is the human mobility in a new city: Transfer mobility knowledge across cities," in *Proceedings of The Web Conference 2020*, 2020, pp. 1355–1365.

[2] C. Rong, J. Ding, Y. Liu, and Y. Li, "A large-scale benchmark dataset for commuting origin-destination matrix generation," *arXiv preprint arXiv:2407.15823*, 2024.

[3] J. Zeng, G. Zhang, C. Rong, J. Ding, J. Yuan, and Y. Li, "Causal learning empowered od prediction for urban planning," in *Proceedings of the 31st ACM international conference on information & knowledge management*, 2022, pp. 2455–2464.